

Graph Harness (SGH) - what got built, and what it showed

A from-scratch Go engine for a paper that shipped no code (arXiv:2604.11378), plus a controlled loop-vs-graph study on real backend code.

90 tests

Go tests passing - 9 packages
2,655 core + 2,872 test LOC

stdlib-only

zero external dependencies
scheduler is -race clean

3 parallel

peak concurrency, one round
10-node DAG - 30 transitions - mock + Gemini

-23%

graph wall-clock vs loop
at scale (the crossover)

2.1x

parallel speedup
245s of work in 117s

0 / 5

fixes left when tokens cut 78%
on a small model

Top row: the engine itself (first-party).

Bottom row: the empirical study (Claude + Gemini, objective pytest ground truth).